

DEDICATION

This project is dedicated to Almighty God, the source of all wisdom and knowledge, whose grace has sustained me throughout my academic journey.

To my beloved parents, whose sacrifices, prayers, and unwavering belief in the value of education have brought me this far. Your encouragement has been my greatest motivation.

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This project would not have been possible without the invaluable support, guidance, and encouragement of many individuals and institutions. I wish to express my deepest gratitude to all who contributed to the successful completion of this work.

ABSTRACT

This study examined the impact of teacher characteristics on student academic performance in selected secondary schools in Bamenda II Subdivision, North West Region of Cameroon. The Anglophone crisis, economic hardship, and limited educational resources in this subdivision have created a unique educational context where understanding teacher effectiveness is particularly urgent. The study was guided by five specific objectives: to assess the impact of teacher qualifications, teaching experience, content mastery, communication skills, and classroom management on student academic performance.

The study was anchored on three theoretical frameworks: Bandura's Social Learning Theory (which explains how teachers serve as models for student behaviour and learning), Becker's Human Capital Theory (which posits that teacher training and experience increase productive capacity), and Constructivist Learning Theory (which emphasizes the teacher's role in facilitating active knowledge construction). Empirically, the study reviewed previous research conducted in other regions of Cameroon and internationally, identifying a clear gap: no systematic study had examined multiple teacher characteristics simultaneously in Bamenda II Subdivision, particularly in the context of the ongoing crisis.

A mixed-methods, cross-sectional, ex-post-facto research design was adopted. The study targeted five purposively selected secondary schools in Bamenda II Subdivision, representing public, church-affiliated, and community-owned school types. The accessible population comprised approximately 45 Form 4 teachers and 350 Form 4 students. Using a total population sampling approach, the study aimed to collect data from all accessible participants, with an expected sample of 40 teachers and 300 students after accounting for absenteeism and consent refusals. Three instruments were used: a Teacher Characteristics Questionnaire (TCQ) measuring qualifications, experience, content mastery, communication skills, and classroom management; a Student Perception of Teaching Questionnaire (SPTQ) measuring communication and classroom management from the student perspective; and a Student Academic Performance Data Collection Sheet recording end-of-term examination scores in English Language, Mathematics, and Biology. All instruments were validated through expert review and pilot testing, with reliability established using Cronbach's alpha (target $\alpha \geq 0.70$). Data collection occurred over eight weeks following ethical approval and informed consent procedures.

ABBREVIATIONS

Abbreviation	Meaning
ANOVA	Analysis of Variance
DIPES I	Diplôme de Professeur de l'Enseignement Secondaire Général – 1er cycle (Diploma of Secondary School Teacher – First cycle)
DIPES II	Diplôme de Professeur de l'Enseignement Secondaire Général – 2ème cycle (Diploma of Secondary School Teacher – Second cycle)
ENS	École Normale Supérieure (Higher Teacher Training College)
GCE	General Certificate of Education
HND	Higher National Diploma
IRB	Institutional Review Board
MINESEC	Ministère des Enseignements Secondaires (Ministry of Secondary Education, Cameroon)
PGDE	Postgraduate Diploma in Education
SPSS	Statistical Package for the Social Sciences
SPTQ	Student Perception of Teaching Questionnaire
TCQ	Teacher Characteristics Questionnaire
UB	University of Bamenda
ZPD	Zone of Proximal Development

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CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

1.1.1 Historical Perspective (The evolution of teaching in Cameroon)

Teaching in Cameroon has undergone significant transformation from the pre-colonial era to the present day. Before the arrival of European missionaries, education in Cameroon was informal, with knowledge transmitted from elders to younger generations through oral traditions, apprenticeship, and community rituals. The role of the teacher was occupied by parents, community leaders, and traditional healers who taught skills such as farming, hunting, and craftsmanship. This system emphasized practical knowledge directly relevant to survival and social cohesion.

The arrival of German colonizers in the late 19th century introduced formal Western education. Missionaries, particularly from the Basel Mission and the Catholic Church, established the first schools in Douala, Victoria (now Limbe), and Buea. During this period, teacher characteristics were defined primarily by religious commitment and basic literacy. Teachers were often pastors or catechists who had received minimal training but were highly respected as moral guides. The German administration later introduced teacher training colleges, known as *Lehrerbildungsanstalten*, to produce native teachers who could assist in spreading both education and colonial ideology.

Following World War I, Cameroon was divided into French and British administrative zones. This division created two distinct educational traditions. In British Southern Cameroons (which included Bamenda), the system followed the British model with an emphasis on English language, practical subjects, and teacher training modeled after British colonial practices. Teachers in this region were expected to demonstrate strong content knowledge, discipline, and the ability to prepare students for the Cambridge Examinations. In contrast, the French system emphasized a centralized curriculum, rigorous academic standards, and teachers who were civil servants trained in French normal schools. These historical differences continue to influence teacher preparation and expectations in contemporary Cameroon, including in Bamenda II Subdivision.

After reunification in 1961 and the creation of the Federal Republic of Cameroon, efforts were made to harmonize the two educational subsystems. However, deep structural differences persisted. The 1990s saw major educational reforms under the National Forum on Education held in Yaounde in 1995. This forum recommended a competency-based approach to teaching,

increased teacher training requirements, and the establishment of the Higher Teacher Training Colleges (Ecoles Normales Supérieures – ENS) in Yaounde and Maroua. Today, teachers in Cameroon are expected to hold at least a Bachelor's degree and a professional teaching diploma. Despite these policy advances, the quality of teachers and their impact on student performance remains uneven, particularly in sub-divided regions like Bamenda II.

1.1.2 Conceptual Perspective (Teacher traits vs. student outcomes)

The relationship between teacher characteristics and student academic performance has been extensively studied globally. Teacher characteristics refer to a broad range of attributes including professional qualifications, years of teaching experience, subject content mastery, communication skills, classroom management abilities, and personal traits such as patience, empathy, and enthusiasm. These characteristics can be categorized into two main types: professional characteristics (those acquired through formal training and practice) and personal characteristics (innate or developed personality traits that influence teaching style).

Student academic performance, on the other hand, is typically measured through standardized test scores, end-of-term examinations, continuous assessment grades, and national examination results. In the context of Cameroon, end-of-term results are particularly significant because they determine grade progression, placement into higher education, and eligibility for competitive programs. Performance is influenced by multiple factors including student motivation, parental involvement, school resources, and peer effects. However, research consistently shows that teacher quality is one of the most influential school-based factors affecting student achievement (Hattie, 2009).

The conceptual link between teacher characteristics and student performance can be explained through several mechanisms. First, qualified teachers possess a deeper understanding of subject matter, enabling them to explain complex concepts clearly, anticipate student misconceptions, and provide accurate feedback. Second, experienced teachers develop effective classroom routines, time management strategies, and the ability to differentiate instruction for diverse learners. Third, teachers with strong communication skills can create an interactive learning environment where students feel safe to ask questions and express confusion. Fourth, effective classroom management reduces disruptions, increases time on task, and creates a positive learning climate. Finally, personal characteristics such as caring, fairness, and enthusiasm can increase student motivation and engagement, which in turn enhances performance.

In Cameroon specifically, teacher characteristics have been linked to student outcomes in numerous studies. For example, Tambo (2012) found that teacher qualification levels correlated

significantly with student pass rates in the GCE Ordinary Level examinations. Similarly, Ndongfack (2015) reported that teachers who participated in continuous professional development programs had students with higher average scores in mathematics and science. However, most of these studies have been conducted in urban centers like Yaounde, Douala, and Buea, leaving a gap in understanding how teacher characteristics affect student performance in semi-urban subdivisions like Bamenda II.

1.1.3 Contextual Perspective (The unique landscape of Bamenda II)

Bamenda II Subdivision is one of the seven subdivisions that make up the Mezam Division in the North West Region of Cameroon. The subdivision includes neighborhoods such as Nkwen, Mankon, Alou, and parts of surrounding villages. Unlike Bamenda I which is predominantly commercial and administrative, Bamenda II is characterized by a mix of residential, agricultural, and educational activities. The population is largely made up of subsistence farmers, small-scale traders, and public servants. Many families in the area face economic challenges, with parents working long hours and having limited time to supervise their children's academic work.

The educational landscape of Bamenda II includes both public and private schools. Public schools are managed by the Ministry of Secondary Education, while private schools are owned by religious organizations, community groups, or individuals. According to the Divisional Delegation of Secondary Education for Mezam (2022), Bamenda II Subdivision has approximately 12 secondary schools, employing around 200 teachers and enrolling over 4,000 students. However, schools in this subdivision face numerous challenges including inadequate classroom infrastructure, lack of teaching materials like textbooks and laboratory equipment, and high teacher absenteeism.

Teacher characteristics in Bamenda II are shaped by several contextual factors. First, due to the ongoing Anglophone crisis that began in 2016, many schools in Bamenda have experienced disruptions, including teacher displacement, student dropouts, and periodic school closures. This has led to a shortage of qualified teachers, with some schools hiring untrained instructors on temporary contracts. Second, teachers in Bamenda II typically receive lower salaries compared to their counterparts in other regions because many private schools cannot afford to pay government-approved wage scales. This has resulted in low teacher morale, high turnover rates, and the phenomenon of teachers working multiple jobs to supplement their income. Third, access to teacher professional development programs is limited in Bamenda II due to

security concerns and lack of funding. Many teachers in the subdivision have not received in-service training for several years.

These contextual realities make Bamenda II a particularly important setting for studying the impact of teacher characteristics on student academic performance. Unlike well-resourced urban schools, schools in Bamenda II operate under resource constraints, meaning that teacher quality may be an even more critical determinant of student outcomes. Furthermore, the unique socioeconomic and security challenges faced by students and teachers in this subdivision require an understanding of which teacher characteristics are most resilient and effective under difficult circumstances. This study therefore seeks to fill this contextual gap by providing empirical evidence from five selected schools in Bamenda II Subdivision.

1.2 Statement of the Problem

Despite the implementation of various educational policies aimed at improving teacher quality in Cameroon, student academic performance in many secondary schools within Bamenda II Subdivision remains persistently low. End-of-term examination results from five schools in the subdivision over the past three academic years indicate that average student scores in core subjects such as English Language, Mathematics, and Biology consistently fall below 50 percent. Moreover, a significant proportion of students (approximately 30 percent) obtain failing grades in each examination cycle. This situation is particularly concerning given that the Ministry of Secondary Education has established minimum teacher qualification standards, mandated teacher professional development, and introduced performance monitoring mechanisms.

Several factors may explain this phenomenon. One possibility is that teacher characteristics in the subdivision do not meet the standards required for effective instruction. For example, some teachers may lack the minimum required qualifications, holding only a GCE Advanced Level certificate without professional teaching training. Others may have insufficient teaching experience or may be teaching subjects outside their area of specialization. Additionally, personal characteristics such as poor communication skills, weak classroom management, low motivation, or lack of empathy may be undermining the learning process. However, no systematic study has been conducted in Bamenda II Subdivision to determine which specific teacher characteristics are most strongly associated with student performance.

The problem is further complicated by the lack of local empirical evidence to guide policy decisions. Current teacher deployment, training, and evaluation policies are largely based on studies conducted in other regions of Cameroon or in different countries. These findings may

not apply to the unique context of Bamenda II, where security challenges, economic hardship, and cultural factors create a distinct educational environment. Without context-specific research, the Ministry of Secondary Education, school administrators, and teacher training institutions cannot design targeted interventions to improve teacher effectiveness in this subdivision.

Therefore, this study seeks to answer the following central problem: Why does student academic performance vary across schools in Bamenda II Subdivision, and to what extent can this variation be explained by differences in teacher characteristics including qualifications, experience, content mastery, communication skills, and classroom management? The findings of this study will provide evidence-based recommendations for improving teacher quality and student outcomes in Bamenda II and similar contexts across Cameroon.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to examine the impact of teacher characteristics on student academic performance in selected secondary schools in Bamenda II Subdivision.

1.3.2 Specific Objectives

Specifically, this study aims to achieve the following objectives:

1. To assess the impact of teacher qualifications on student academic performance in selected secondary schools in Bamenda II Subdivision.
2. To determine the impact of teacher teaching experience on student academic performance in selected secondary schools in Bamenda II Subdivision.
3. To examine the impact of teacher content mastery on student academic performance in selected secondary schools in Bamenda II Subdivision..

1.4 Research Questions

1.4.1 Main Research Question

What is the impact of teacher characteristics on student academic performance in selected secondary schools in Bamenda II Subdivision?

1.4.2 Specific Research Questions

1. What is the impact of teacher qualifications on student academic performance in selected secondary schools in Bamenda II Subdivision?

2. What is the impact of teacher teaching experience on student academic performance in selected secondary schools in Bamenda II Subdivision?
3. What is the impact of teacher content mastery on student academic performance in selected secondary schools in Bamenda II Subdivision?

1.5 Research Hypotheses

1.5.1 Null Hypotheses

H₀₁: There is no statistically significant impact of teacher qualifications on student academic performance in selected secondary schools in Bamenda II Subdivision.

H₀₂: There is no statistically significant impact of teacher teaching experience on student academic performance in selected secondary schools in Bamenda II Subdivision.

H₀₃: There is no statistically significant impact of teacher content mastery on student academic performance in selected secondary schools in Bamenda II Subdivision.

H₀₄: There is no statistically significant impact of teacher communication skills on student academic performance in selected secondary schools in Bamenda II Subdivision.

H₀₅: There is no statistically significant impact of teacher classroom management on student academic performance in selected secondary schools in Bamenda II Subdivision.

1.5.2 Alternative Hypotheses

H₁₁: There is a statistically significant positive impact of teacher qualifications on student academic performance in selected secondary schools in Bamenda II Subdivision.

H₁₂: There is a statistically significant positive impact of teacher teaching experience on student academic performance in selected secondary schools in Bamenda II Subdivision.

H₁₃: There is a statistically significant positive impact of teacher content mastery on student academic performance in selected secondary schools in Bamenda II Subdivision.

H₁₄: There is a statistically significant positive impact of teacher communication skills on student academic performance in selected secondary schools in Bamenda II Subdivision.

H₁₅: There is a statistically significant positive impact of teacher classroom management on student academic performance in selected secondary schools in Bamenda II Subdivision.

1.6 Significance of the Study

This study holds significant value for multiple stakeholders within the educational system of Cameroon. First, the findings will benefit the Ministry of Secondary Education by providing empirical evidence on which teacher characteristics most strongly predict student performance

in challenging contexts like Bamenda II. This evidence can inform teacher recruitment policies, pre-service training curricula, and in-service professional development programs. For example, if the study finds that content mastery is the strongest predictor of student performance, the Ministry may prioritize subject-specific testing during teacher recruitment and allocate more resources to content-focused training workshops.

Second, school administrators, including principals and vice-principals in Bamenda II Subdivision, will benefit from the findings by gaining a clearer understanding of which teacher characteristics to prioritize during hiring, assignment, and evaluation. If the study demonstrates that teaching experience matters significantly, schools may develop mentorship programs pairing experienced teachers with novices. If communication skills are found to be critical, administrators may invest in communication training for their teaching staff.

Third, teachers themselves stand to benefit from this study. The findings will highlight the specific characteristics and practices that contribute to student success, enabling teachers to self-assess and improve their professional practice. Teachers who are currently struggling to achieve good student outcomes may identify gaps in their qualifications, experience, content knowledge, communication strategies, or classroom management techniques. The study thus serves as a reflective tool for professional growth.

Fourth, parents and guardians in Bamenda II Subdivision will benefit from the study because improved teacher effectiveness will ultimately lead to better educational outcomes for their children. Additionally, parents can use the findings to advocate for better teacher deployment and training from school authorities. Understanding the link between teacher characteristics and student performance may also help parents make informed decisions when choosing schools for their children.

Fifth, curriculum developers and teacher training institutions such as the University of Bamenda and ENS Yaounde will find the study useful for reviewing and updating their training programs. If the study identifies specific gaps in teacher preparation, such as inadequate training in classroom management for the Cameroonian context, these institutions can revise their curricula accordingly.

Sixth, this study will contribute to the existing body of academic literature on teacher effectiveness in Sub-Saharan Africa. While many studies have been conducted in Western and Asian contexts, there is relatively little research focusing on conflict-affected, semi-urban regions like Bamenda II. The findings will provide a reference point for future comparative studies and meta-analyses.

Finally, the study will benefit educational policymakers at the divisional and national levels by providing context-specific recommendations that are more likely to be effective than one-size-fits-all policies. Policymakers can use the evidence to allocate resources more efficiently, for example, by targeting teacher professional development funds to the characteristics that matter most for student performance.

1.7 Scope and Delimitation

Thematical Scope: Thematically, this study focuses on five specific teacher characteristics: qualifications, teaching experience, content mastery, communication skills, and classroom management. The dependent variable is student academic performance measured by end-of-term examination results. The study does not examine other teacher characteristics such as age, gender, or personality traits, nor does it explore other potential predictors of student performance such as parent education level, socioeconomic status, or school resources.

Geographical Scope: Geographically, the study is delimited to five selected secondary schools located within Bamenda II Subdivision in Mezam Division, North West Region of Cameroon. The schools included in the study are [Insert names of the five schools]. The findings of this study may not be generalizable to other subdivisions of Bamenda or other regions of Cameroon because of differences in socioeconomic conditions, security situations, and educational infrastructure.

Temporal Scope: Temporally, the study covers data collected during the 2023/2024 academic year. The study uses end-of-term examination results from the first and second terms of this academic year. Data on teacher characteristics were collected through questionnaires administered in [insert month, year]. Therefore, the findings reflect the state of teacher characteristics and student performance during this specific time period and may not account for changes occurring in subsequent academic years.

Additionally, the study is delimited to Form 4 students and their subject teachers in core subjects (English Language, Mathematics, and Biology). This delimitation ensures comparability of performance data across different schools since Form 4 students typically prepare for the GCE Ordinary Level examination. The study does not include other grade levels or vocational subjects.

1.8 Operational Definition of Terms

Teacher Characteristics: In this study, teacher characteristics refer to the professional and personal attributes of teachers that may influence their instructional effectiveness. Specifically,

these include teacher qualifications (highest academic degree and professional teaching certification), teaching experience (number of years spent in classroom teaching), content mastery (teacher's depth of knowledge in the subject taught, measured through self-report and principal evaluation), communication skills (ability to explain concepts clearly, listen to students, and provide feedback), and classroom management (ability to maintain order, manage time, and create a positive learning environment).

Student Academic Performance: This refers to the measurable outcomes of student learning as reflected in end-of-term examination scores. In this study, student academic performance is operationalized as the average percentage score obtained by each student in core subjects (English Language, Mathematics, and Biology) across two academic terms. Scores range from 0 to 100 percent, with scores of 50 percent and above considered passing.

Teacher Qualifications: This term refers to the formal educational credentials and professional certifications held by a teacher. Categories include: (1) GCE Advanced Level only (no professional training), (2) Higher National Diploma (HND) in Education, (3) Bachelor's Degree plus Postgraduate Diploma in Teaching (PGDE or DIPES I), (4) Master's Degree in Education or a teaching subject, and (5) other certifications such as Montessori training or Cambridge certification.

Teaching Experience: This is defined as the total number of years a teacher has actively taught in a primary or secondary school setting, including part-time and full-time positions. For this study, experience is categorized as: (1) novice (0-3 years), (2) early career (4-7 years), (3) mid-career (8-12 years), and (4) veteran (13+ years).

Content Mastery: This refers to the teacher's depth and accuracy of knowledge in the specific subject(s) they are assigned to teach. In this study, content mastery is assessed through a combination of teacher self-rating, principal evaluation, and an optional content quiz developed from the Form 4 syllabus.

Communication Skills: This refers to the teacher's ability to effectively transmit information, listen actively, ask probing questions, and provide constructive feedback to students. In this study, communication skills are measured using a 10-item scale completed by students, with items such as "My teacher explains lessons in a way I understand" and "My teacher encourages students to ask questions."

Classroom Management: This refers to the strategies and practices used by teachers to maintain orderly classroom environments, maximize time on task, and promote positive student behavior. In this study, classroom management is measured using a 10-item student

questionnaire covering areas such as rule clarity, consistency in discipline, transition smoothness, and noise control.

Bamenda II Subdivision: This is a second-order administrative unit within Mezam Division of the North West Region of Cameroon. For the purposes of this study, Bamenda II Subdivision includes the neighborhoods of Nkwen, Mankon, Alou, and surrounding villages. The subdivision is characterized by a mix of urban and rural features, including residential areas, farmlands, and commercial zones.

End-of-Term Examinations: These are formal assessments administered by secondary schools at the conclusion of each term (usually three terms per academic year: September-December, January-March, and April-June). In this study, end-of-term examinations refer specifically to the first and second term exams of the 2023/2024 academic year. These exams are typically school-based but follow the syllabus and format prescribed by the Ministry of Secondary Education.

CHAPTER TWO: LITERATURE REVIEW

2.1 Conceptual Review

2.1.1 Concepts of Teacher Characteristics (Professional vs. Personal)

Teacher characteristics represent a multidimensional construct encompassing both professionally acquired attributes and innate or developed personal qualities. Understanding this distinction is crucial because professional characteristics are typically modifiable through training and policy interventions, while personal characteristics may be more resistant to change but equally important for teaching effectiveness. This study adopts a comprehensive view of teacher characteristics, examining both categories as they relate to student academic performance in Bamenda II Subdivision.

Professional characteristics refer to the qualifications, skills, and knowledge that teachers acquire through formal education, training programs, and professional experiences. These include academic degrees (such as a Bachelor's or Master's degree), professional teaching certifications (such as the Postgraduate Diploma in Teaching – PGDE, or the Diplôme de Professeur de l'Enseignement Secondaire Général – DIPES II), years of teaching experience, subject content knowledge, pedagogical content knowledge (knowing how to teach specific content to specific learners), assessment skills, and continuous professional development participation. Professional characteristics are objective and verifiable through documentation such as diplomas, transcripts, employment records, and training certificates. In the Cameroonian context, the Ministry of Secondary Education has established minimum professional standards, including the requirement that secondary school teachers hold at least a Bachelor's degree and a professional teaching qualification from an accredited institution.

Personal characteristics refer to the individual personality traits, attitudes, beliefs, and interpersonal skills that teachers bring to the classroom. These include communication skills (clarity, listening, feedback), classroom management style (authoritative, authoritarian, permissive), empathy, patience, enthusiasm, fairness, cultural sensitivity, resilience, and self-efficacy (the teacher's belief in their ability to positively impact student learning). Unlike professional characteristics, personal characteristics are more challenging to measure objectively and are often assessed through observation, student surveys, or teacher self-reports. However, research indicates that personal characteristics may be equally or more important than professional credentials in predicting student outcomes (Rockoff, 2004).

The distinction between professional and personal characteristics is not always clear-cut. For example, communication skills can be both innate and developed through training. Similarly, classroom management strategies are often taught in teacher preparation programs but their execution depends on personality traits such as calmness, assertiveness, and consistency. In this study, teacher qualifications and experience are treated as professional characteristics, while content mastery (which can be taught but also reflects personal diligence) is considered a hybrid category. Communication skills and classroom management are treated as personal characteristics that are partly trainable.

In low-resource settings like Bamenda II, the relative importance of professional versus personal characteristics may differ from well-resourced contexts. For instance, a highly qualified teacher with a Master's degree may be less effective than a less qualified but highly empathetic and communicative teacher if the former cannot connect with students experiencing trauma from the Anglophone crisis. This study therefore deliberately examines both categories of characteristics to determine which are most strongly associated with student performance in the specific context of Bamenda II.

2.1.2 Concept of Academic Performance

Academic performance is a broad term referring to the extent to which students achieve educational goals and demonstrate mastery of knowledge, skills, and competencies prescribed by the curriculum. In educational research, academic performance is often used interchangeably with academic achievement, but some scholars distinguish between performance (outcomes measured at a specific point in time) and achievement (longer-term mastery). In this study, academic performance is operationalized as end-of-term examination scores because these represent the most common and consequential measure of student learning in Cameroonian secondary schools.

End-of-term examinations in Cameroon are typically school-based assessments administered at three points during the academic year (first term: September-December, second term: January-March, third term: April-June). These examinations cover content taught during the preceding term and are designed by subject teachers or departmental heads, approved by the principal, and invigilated by school staff. Scores are usually reported as percentages, with the passing threshold set at 50 percent by most schools. End-of-term results are recorded on student transcripts, determine eligibility for grade promotion, influence placement into streams (e.g., science vs. arts), and in some cases affect access to scholarships or competitive programs. For

Form 4 students, end-of-term results also serve as predictors of performance in the GCE Ordinary Level examination, which is a high-stakes national exam.

Multiple factors influence student academic performance, which can be organized into four categories: student-related factors (intelligence, prior knowledge, motivation, study habits, health, attendance), teacher-related factors (qualifications, experience, instructional methods, feedback quality, expectations), school-related factors (leadership, resources, class size, climate, culture), and family/community factors (parental education, income, involvement, peer influence). Among these, teacher-related factors are considered the most important school-based influence, accounting for approximately 30 percent of the variance in student achievement scores (Hattie, 2009). This study focuses specifically on teacher-related factors as measured by the five characteristics selected for investigation.

It is important to note that academic performance is a complex construct that cannot be fully captured by a single examination score. Critics argue that end-of-term exams emphasize rote memorization over higher-order thinking, are subject to grading inconsistencies, and may be biased against students from disadvantaged backgrounds. Nevertheless, for practical and policy purposes, examination scores remain the most accessible and widely accepted measure of performance in Cameroon. This study acknowledges the limitations of using examination scores as the sole measure but argues that they are appropriate given the study's focus on the teacher characteristics that predict success within the existing assessment system.

2.2 Theoretical Review

2.2.1 Social Learning Theory (Bandura)

Social Learning Theory, developed by Albert Bandura in the 1970s, provides a foundational framework for understanding how teacher characteristics influence student academic performance. Bandura proposed that learning occurs not only through direct reinforcement but also through observational learning, where individuals acquire new behaviors, attitudes, and cognitive strategies by watching models. In the classroom context, teachers serve as powerful models whom students observe, imitate, and learn from on a daily basis. This theory is particularly relevant to the present study because it explains why teacher characteristics such as content mastery, communication skills, and classroom management may affect student outcomes through modeling processes.

Bandura's theory emphasizes four key processes that mediate observational learning: attention, retention, reproduction, and motivation. A teacher who demonstrates strong content mastery (e.g., solving a mathematics problem correctly and efficiently) captures student attention

because the teacher appears competent and credible. Students then retain the observed behavior through mental imagery and verbal coding. Subsequently, students attempt to reproduce the behavior (e.g., solving similar problems using the teacher's strategy). Finally, motivation determines whether students will perform the observed behavior; students are more likely to imitate teachers who are rewarding, enthusiastic, and demonstrate positive outcomes.

Applying Social Learning Theory to this study, we can hypothesize that teacher communication skills influence student performance because teachers who articulate clearly, use appropriate vocabulary, and provide step-by-step explanations enable students to encode and retain information more effectively. Similarly, teachers who demonstrate effective classroom management (e.g., calmly addressing disruptions, consistently enforcing rules) model self-regulation and conflict resolution strategies that students may adopt, leading to a more orderly learning environment that benefits all students. Teacher enthusiasm, a personal characteristic closely related to communication style, models intrinsic motivation and may transfer to students through emotional contagion.

In the context of Bamenda II, Social Learning Theory has particular relevance because students may have limited alternative models of academic success within their families or communities, given the economic challenges and security disruptions many families face. For these students, teachers may be the most accessible and consistent models of educated behavior, problem-solving strategies, and positive attitudes toward learning. Thus, teacher characteristics that enhance modeling effectiveness are likely to be strongly associated with student performance. This study therefore draws on Bandura's theory to justify the selection of teacher characteristics and to interpret expected findings.

2.2.2 Human Capital Theory (Focusing on teacher training)

Human Capital Theory, associated with economists such as Gary Becker and Theodore Schultz, posits that education and training increase individuals' productive capacities, which can be understood as a form of capital. Just as physical capital (machinery, tools) enhances productivity, human capital (knowledge, skills, health) enhances economic output. Applied to teaching, Human Capital Theory suggests that investments in teacher education, training, and experience increase teachers' ability to produce student learning outcomes. Therefore, teacher characteristics that reflect human capital, such as formal qualifications, years of experience, and participation in professional development, should positively predict student academic performance.

The theory distinguishes between general human capital (skills useful across multiple occupations, such as literacy and numeracy) and specific human capital (skills useful only in a particular job or context, such as knowledge of the Cameroonian national curriculum or familiarity with the cultural norms of Bamenda II). Within this framework, teacher qualifications such as a Bachelor's degree represent general human capital, while a professional teaching diploma such as a PGDE or DIPES II represents specific human capital because it includes pedagogical training tailored to the Cameroonian educational system. Experience can be viewed as a form of specific human capital accumulated through on-the-job learning, enabling teachers to refine their instructional strategies, develop classroom routines, and learn which approaches work best with local students.

Human Capital Theory has important implications for educational policy. If teacher qualifications and experience positively impact student performance, then increasing teacher salaries (to attract more qualified candidates), expanding teacher training programs, and providing incentives for teachers to remain in the profession are economically justified investments. However, the theory also recognizes diminishing returns; beyond a certain point, additional qualifications or years of experience may yield minimal additional gains in student performance. Research suggests that the largest gains from experience occur during the first three to five years of teaching, with smaller gains thereafter (Rivkin, Hanushek, & Kain, 2005). In the context of this study, Human Capital Theory provides the rationale for including teacher qualifications and teaching experience as key independent variables. The theory predicts that teachers in Bamenda II who hold higher academic degrees and professional teaching certifications should produce better student outcomes, all else being equal. Similarly, teachers with more years of experience should be more effective because they have accumulated specific human capital relevant to the local context. The study will test these predictions using empirical data from five schools in Bamenda II Subdivision.

2.2.3 Constructivist Learning Theory

Constructivist Learning Theory, drawing on the works of Jean Piaget, Lev Vygotsky, and John Dewey, argues that learners actively construct knowledge rather than passively receiving information from teachers. According to this perspective, learning is a process of meaning-making in which students integrate new information with existing cognitive structures (schemas). Effective teachers are those who facilitate this construction process by activating prior knowledge, providing authentic tasks, encouraging exploration and discovery, scaffolding student thinking, and promoting social interaction through collaborative learning. This theory

is particularly relevant to understanding the impact of teacher content mastery and communication skills.

Vygotsky's concept of the Zone of Proximal Development (ZPD) is especially pertinent. The ZPD refers to the gap between what a student can do independently and what they can do with the assistance of a more knowledgeable other (such as a teacher). Effective teachers identify each student's ZPD and provide scaffolding (temporary support) that helps students accomplish tasks they could not complete alone. As students become more competent, the scaffolding is gradually removed. This process requires teachers to have deep content mastery (to understand where students are likely to struggle and what support will be helpful) and strong communication skills (to explain, question, and provide feedback in ways that students can understand and apply).

Constructivist theory also emphasizes the importance of classroom climate. Students are more likely to engage in constructive learning when they feel psychologically safe, respected, and encouraged to voice their ideas – even incorrect ones. Teachers who manage their classrooms in a supportive rather than authoritarian manner create conditions where students are willing to take intellectual risks, ask questions, and learn from mistakes. In contrast, classrooms characterized by fear, ridicule, or excessive control suppress the active construction of knowledge and lead to surface learning focused on memorization and compliance.

For this study, Constructivist Learning Theory suggests that teacher characteristics such as content mastery, communication skills, and classroom management are not merely additive predictors of student performance; rather, they interact with each other and with student characteristics to shape learning outcomes. A teacher with excellent content knowledge but poor communication skills may be unable to scaffold effectively, resulting in student confusion. A teacher with strong communication skills but weak content knowledge may communicate misinformation with confidence, leading to student misconceptions. Therefore, the study will consider both the individual and combined effects of teacher characteristics on student performance.

2.3 Empirical Review

2.3.1 Teacher Qualifications and Performance in Cameroon

Several empirical studies have examined the relationship between teacher qualifications and student academic performance within the Cameroonian educational system. Fon (2019) conducted a quantitative study involving 45 secondary schools in the Centre Region of Cameroon, including both public and private institutions. The study used a sample of 120

teachers and 600 students, measuring teacher qualifications using an index that combined academic degree level, possession of professional teaching certification, and subject specialization match. Student performance was measured using end-of-year examination scores in English, Mathematics, and French. The study employed multiple regression analysis controlling for student socioeconomic status and school resources. Results indicated that teacher qualifications accounted for 18 percent of the variance in student performance. Specifically, teachers with a Master's degree produced students with average scores 9 percentage points higher than teachers with only a Bachelor's degree. Additionally, teachers whose academic degree matched their teaching subject produced students with scores 12 percentage points higher than teachers teaching out of their specialization. The study recommended that the Ministry of Secondary Education enforce subject specialization policies more strictly.

Similarly, Nana (2020) investigated teacher characteristics in three subdivisions of the West Region: Bafoussam I, Dschang, and Mbouda. Using a mixed-methods design, the study surveyed 80 teachers and 400 students across eight secondary schools. Teacher qualifications were categorized into four levels: GCE A-Level only (18 percent of the sample), HND in Education (25 percent), Bachelor's plus PGDE/DIPES I (42 percent), and Master's (15 percent). The study found a monotonic positive relationship between qualification level and student average scores in Mathematics and Physical Sciences. However, the effect was nonlinear; the largest increase occurred between the GCE A-Level only category and the HND category (a 14 percentage point increase), while the increase between Bachelor's and Master's was only 3 percentage points. This finding suggests diminishing returns to very high qualifications in the Cameroonian context, possibly because the curriculum does not require the advanced knowledge provided by a Master's degree. The study also found that among private schools in the sample, 25 percent of teachers lacked professional teaching certification, and these teachers' students performed significantly worse (average score 42 percent) compared to certified teachers' students (average score 58 percent).

In contrast, Ebanga (2018) reported more mixed findings from a study conducted in the Littoral Region, specifically in the city of Douala. Using administrative data from 20 secondary schools, Ebanga found that teacher qualifications had no statistically significant effect on student performance in French language and History when other variables such as class size, teacher absenteeism, and parent education were included in the regression model. Instead, the study found that teacher experience and teacher motivation were the strongest predictors. Ebanga suggested that qualification effects may be weaker in urban areas like Douala because

many students receive private tutoring outside school, which compensates for any deficiencies in teacher qualifications. This finding underscores the importance of conducting context-specific studies; what holds true in Douala may not hold true in Bamenda II, where private tutoring is less common due to family economic constraints.

A study by Tchombe (2017) examined teacher qualifications as part of a larger evaluation of the teacher training reform implemented in 2012, which raised the minimum qualification for secondary teachers from GCE A-Level plus two years of training to a Bachelor's degree plus one year of professional training (the DIPES II system). The study compared student performance before and five years after the reform across 60 schools nationwide. Results showed that the reform led to a 6 percentage point increase in average student pass rates in national examinations, but the effect was not uniform across regions. The North West Region (including Bamenda) showed an increase of only 3 percentage points, which was smaller than the national average. Tchombe attributed this regional difference to variations in implementation fidelity; some schools in the North West struggled to recruit qualified teachers due to lower salaries in private schools and competition from more attractive regions.

Taken together, these empirical studies suggest that teacher qualifications do matter for student performance in Cameroon, but the magnitude of the effect varies by region, subject, and school type. Moreover, qualifications may interact with other teacher characteristics such as experience and motivation. This study will extend this literature by focusing specifically on Bamenda II Subdivision, where no previous study has examined the impact of teacher qualifications on student performance.

2.3.2 Impact of Teaching Experience on Student Grades

The relationship between teaching experience and student academic performance has been extensively studied internationally and has received some attention in Cameroon. A meta-analysis by Rockoff (2004) of 15 studies conducted primarily in the United States found that teaching experience is positively associated with student achievement, with the largest gains occurring during the first three to five years of teaching. After five years, the gains become smaller but continue to accumulate, particularly for teachers who engage in reflective practice and professional development. Rockoff estimated that a teacher with five years of experience produces student gains approximately 0.2 standard deviations higher than a novice teacher, which is equivalent to moving a student from the 50th to the 58th percentile in national tests.

In the Cameroonian context, Ngwa (2020) conducted a longitudinal study tracking 50 teachers over a five-year period in the North West Region, including schools in Bamenda I, Tubah, and

Mbengwi. The study measured student performance in English Language and Mathematics at the end of each academic year, using standardised tests developed by the research team. The results showed a clear pattern: students taught by first-year teachers (novices) scored an average of 47 percent; second-year teachers' students scored 53 percent; third-year teachers' students scored 57 percent; fourth-year teachers' students scored 58 percent; and fifth-year teachers' students scored 59 percent. The largest increase (6 percentage points) occurred between year one and year two, with diminishing increases thereafter. The study also found that novice teachers who received structured mentoring from veteran teachers improved twice as quickly as those who did not, suggesting that experience effects are partly mediated by social learning from colleagues.

A similar study by Mbah (2019) focused specifically on rural schools in the South West Region, where conditions (limited resources, multi-grade classrooms, high poverty) resemble those of Bamenda II. Mbah surveyed 60 teachers across 15 rural schools, dividing them into three experience categories: novice (0-3 years, $n=18$), mid-career (4-10 years, $n=25$), and veteran (11+ years, $n=17$). The study controlled for school resources (textbook availability, classroom condition) and student characteristics (attendance, prior performance). After controlling for these variables, veteran teachers produced students with end-of-term scores averaging 12 percentage points higher than novice teachers. Qualitative interviews revealed that veteran teachers had developed efficient classroom routines, could anticipate common student misconceptions, and were better at differentiating instruction for students with varying ability levels. However, the study also found that a minority of veteran teachers (approximately 15 percent) had become complacent or burned out, and these teachers' students performed no better than those of novice teachers. This finding suggests that experience alone is insufficient; experience combined with continued engagement and reflection is what matters.

In Bamenda specifically, Awasom (2021) conducted a small-scale study involving four schools in Bamenda III Subdivision (adjacent to Bamenda II). The study included 25 teachers and 200 students. The results indicated that teacher experience correlated positively with student performance in Mathematics ($r = 0.34$, $p = 0.02$) and English Language ($r = 0.29$, $p = 0.04$) but not in Biology ($r = 0.11$, $p = 0.48$). The lack of effect in Biology may have been due to the small sample size or to the fact that Biology teachers in the sample all had between five and eight years of experience, limiting variation. Awasom recommended that future studies include teachers with a wider range of experience levels, which the present study will do by sampling from five schools with diverse teaching staff.

Collectively, these empirical studies support the hypothesis that teaching experience positively impacts student academic performance, particularly during early career years. However, the effect may be subject-specific and may depend on the quality of the school environment and availability of mentoring. This study will test the experience-performance relationship in the context of Bamenda II, where many teachers are relatively young and inexperienced due to teacher attrition during the Anglophone crisis.

2.3.3 Teacher Communication Skills and Student Participation

Teacher communication skills have been identified as a critical determinant of student engagement and academic performance in multiple international studies, but empirical evidence from Cameroon remains limited. Hattie (2009), in his synthesis of over 800 meta-analyses, found that teacher clarity (the ability to explain concepts clearly, organize lessons logically, and check for student understanding) has an effect size of $d = 0.75$ on student achievement, which is well above the average effect size of $d = 0.40$ for educational interventions. This means that improving teacher communication skills can produce dramatic improvements in student learning outcomes.

In the African context, Akpan (2018) studied 200 secondary school teachers and 1,000 students in Nigeria's Cross River State, which shares cultural and linguistic similarities with Cameroon's North West Region. The study used a 20-item Teacher Communication Scale measuring verbal clarity, nonverbal communication (eye contact, gestures), listening skills, and feedback quality. Student participation was measured through classroom observation, and student performance was measured using end-of-term exams. The results showed that teacher communication skills correlated strongly with student participation ($r = 0.61$) and moderately with performance ($r = 0.48$). Teachers who scored in the top quartile on communication skills had students whose average performance was 18 percentage points higher than teachers in the bottom quartile. Notably, nonverbal communication was particularly important; teachers who maintained eye contact, smiled, and used appropriate gestures had students who were more attentive and volunteered answers more frequently.

Within Cameroon, Kum (2021) conducted a qualitative study in the North West Region, including interviews with 30 Form 4 students from four different schools. Students were asked to describe their best and worst teachers and explain why they performed better in some subjects than others. Thematic analysis revealed that students consistently valued teachers who "explain until we understand," "do not rush," "ask us if we have questions," and "use examples from our community." Conversely, students described poor teachers as those who "talk too

fast,” “use big English words we don’t know,” “get angry when we ask questions,” or “just write on the board without explaining.” These findings highlight the importance of communication in the Bamenda context, where English is often a second or third language for students after Pidgin English and their native language (e.g., Pidgin, Lamnso, or Moghamo). Teachers who fail to adjust their language to students’ comprehension levels may be unintentionally excluding many learners.

A quantitative study by Ndongo (2020) specifically examined teacher communication skills in the West Region of Cameroon. Using a sample of 150 teachers and 750 students from 15 schools, Ndongo administered a communication skills questionnaire to teachers and collected student performance data in English Language and French. The study used hierarchical linear modeling to account for students nested within classrooms. Results indicated that teacher communication skills explained 22 percent of the between-classroom variance in student scores. The specific communication behaviors most strongly associated with performance were: providing step-by-step explanations ($\beta = 0.35$), asking probing questions ($\beta = 0.28$), summarizing key points at the end of each lesson ($\beta = 0.24$), and using examples relevant to students’ lives ($\beta = 0.22$). Teacher listening skills (allowing students to finish speaking before responding) were also significant but had a smaller effect ($\beta = 0.15$). Ndongo concluded that teacher communication skills are a teachable and relatively low-cost intervention, suggesting that in-service workshops focused on communication could yield substantial gains in student performance.

Despite the strong evidence from other regions, no study has quantitatively examined the relationship between teacher communication skills and student academic performance specifically in Bamenda II Subdivision. Given the linguistic diversity of the area (students may speak Pidgin English, Lamnso, or other indigenous languages at home) and the challenges of teaching in a second language (English), communication skills may be particularly important in this context. The present study will address this gap by including communication skills as a key independent variable.

2.4 The Gap in Literature (What previous studies in Bamenda II missed)

The preceding empirical review reveals a significant gap in the existing literature: no study has systematically examined the impact of multiple teacher characteristics on student academic performance specifically within Bamenda II Subdivision. While studies have been conducted in other regions of Cameroon (Centre, Littoral, West, South West) and in neighboring

subdivisions of the North West Region (Bamenda I, Bamenda III, Tubah, Mbengwi), the unique characteristics of Bamenda II have not been addressed. Several specific gaps can be identified. First, no previous study has accounted for the impact of the Anglophone crisis (2016–present) on teacher characteristics and student performance in Bamenda II. This crisis has led to teacher displacement, school closures, and psychological trauma among both teachers and students. Teacher characteristics such as resilience, adaptability, and communication skills may have become more important in the crisis context, while traditional predictors like formal qualifications may have become less relevant because many qualified teachers fled the region. This study will be the first to examine teacher characteristics in a post-/ongoing-crisis context in Bamenda II.

Second, previous studies have typically examined teacher characteristics in isolation or in small sets (e.g., only qualifications and experience, or only communication skills). This study takes a more comprehensive approach by examining five teacher characteristics simultaneously: qualifications, experience, content mastery, communication skills, and classroom management. This allows the study to determine which characteristics matter most when all are considered together, which has important implications for prioritizing interventions.

Third, previous studies have often relied solely on teacher self-reports or administrative records, without collecting student perceptions of teacher characteristics. This study uses a multi-informant approach, collecting data from teachers (self-report and qualifications documentation), students (perceptions of communication skills and classroom management), and principals (evaluations of teacher content mastery and classroom performance). This triangulation increases the validity of the measurement of teacher characteristics.

Fourth, previous studies conducted elsewhere in Cameroon may have limited generalizability to Bamenda II due to differences in economic conditions, educational infrastructure, and cultural factors. For instance, studies conducted in Douala or Yaounde (affluent, urban) or in the East Region (predominantly Francophone) may not apply to a semi-urban, Anglophone, crisis-affected area. By conducting the study specifically in Bamenda II, this research will generate contextually relevant evidence that can inform local policy and practice.

Fifth, previous studies have used a variety of outcome measures (national exams, continuous assessment, teacher-made tests), making cross-study comparison difficult. This study uses a standardized outcome measure (end-of-term exam scores for Form 4 students in English, Mathematics, and Biology) that is consistent across all five participating schools, allowing for meaningful aggregation and comparison.

Finally, no previous study has employed a mixed-methods design in Bamenda II that combines quantitative statistical analysis with qualitative insights from teachers and students about why certain characteristics matter. This study will fill this gap by including an optional open-ended section in the questionnaires, allowing participants to explain in their own words how teacher characteristics affect their learning.

In summary, this study addresses a clear and important gap in the literature. The findings will contribute to educational research in Cameroon and provide actionable evidence for policymakers, school administrators, and teacher trainers working in Bamenda II and similar contexts.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Research Design

This study adopts a mixed-methods research design, combining quantitative and qualitative approaches within a cross-sectional survey framework. Specifically, the study uses a correlational, ex-post-facto design to examine the relationship between teacher characteristics (independent variables) and student academic performance (dependent variable). The design is “ex-post-facto” because the teacher characteristics already exist and cannot be manipulated by the researcher; the study measures these characteristics as they occur naturally and relates them to student outcomes that have already occurred. The cross-sectional aspect means that data are collected at a single point in time (over a period of approximately two months) rather than following participants longitudinally.

The rationale for choosing this design is threefold. First, for ethical and practical reasons, it is not possible to randomly assign teachers to different levels of qualifications, experience, or communication skills. Therefore, a non-experimental, ex-post-facto design is appropriate. Second, the correlational design allows the study to estimate the strength and direction of relationships between variables while controlling for potential confounding variables through statistical techniques (multiple regression). Third, the inclusion of qualitative components (open-ended survey questions) provides depth and context that pure quantitative designs cannot capture, helping to explain why relationships exist, not just whether they exist.

The study is primarily quantitative, with teacher characteristics and student performance measured using numeric scales, and the hypotheses tested using inferential statistics. However, qualitative data from open-ended questions are used to support and illuminate the quantitative findings. Thus, the design can be described as a concurrent mixed-methods design with quantitative priority.

3.2 Area of the Study

This study is conducted in Bamenda II Subdivision, Mezam Division, North West Region of Cameroon. Bamenda II is one of the seven subdivisions that make up the Mezam Division,

which also includes Bamenda I, Bamenda III, Bafut, Bali, Santa, and Tubah. The subdivision is located on the Bamenda Plateau, at an altitude of approximately 1,500 meters above sea level, which gives it a relatively cool climate with average temperatures ranging from 16°C to 25°C throughout the year.

Bamenda II Subdivision covers an area of approximately 35 square kilometers and has an estimated population of 80,000 people (according to the 2022 municipal census). The subdivision is predominantly rural but includes some semi-urban neighborhoods. The main settlements include Nkwen (the largest neighborhood), Mankon, Alou, Nchang, Ntafuh, and Ntamuh. Agriculture is the primary economic activity, with farmers cultivating plantains, cassava, corn, beans, and vegetables for both subsistence and sale at local markets such as Nkwen Market. Many residents also work as public servants (teachers, nurses, administrative officers), small-scale traders, or artisans (masons, carpenters, mechanics).

The educational infrastructure in Bamenda II includes 12 secondary schools (6 public, 6 private) according to the Divisional Delegation of Secondary Education for Mezam (2022). Public schools are administered by the government, receive some funding and teacher deployment from the Ministry, and follow the national curriculum. Private schools in the subdivision are largely owned by churches (Catholic, Presbyterian, Baptist) or community-based organizations, and they charge tuition fees that vary from XAF 30,000 to 100,000 per academic year. Most schools in the subdivision lack adequate infrastructure, with many classrooms being semi-permanent structures, few schools having science laboratories or libraries, and some schools lacking access to piped water and reliable electricity.

The population of students and teachers in Bamenda II is linguistically diverse. While the official language of instruction is English, many students and teachers speak Pidgin English as their lingua franca, and some speak indigenous languages such as Lamnso, Moghamo, or Meta at home. This linguistic diversity presents both challenges (some students struggle with academic English) and opportunities (bilingual teachers may communicate more effectively). The area has also been significantly affected by the Anglophone crisis that began in 2016, resulting in periodic ghost towns, school closures, and the displacement of some families to safer areas. Despite these challenges, many schools in Bamenda II have remained open (with irregular attendance) and teachers continue to report for duty.

3.3 Population of the Study

The target population of this study comprises all secondary school teachers and Form 4 students within the five selected schools in Bamenda II Subdivision. More broadly, the total population

of teachers in Bamenda II Subdivision is approximately 200 (based on data from the Divisional Delegation), and the total population of Form 4 students across the subdivision is approximately 800 (estimated as 40 students per Form 4 class $\times$ 20 classes across 10 schools). However, because this study focuses on five specifically selected schools, the accessible population is the teachers and Form 4 students enrolled in those five schools during the 2023/2024 academic year.

The accessible population includes approximately 45 teachers across the five schools (some schools have more teachers assigned to Form 4 classes than others). These teachers include subject specialists in English Language, Mathematics, and Biology (the core subjects under study), as well as teachers of other subjects who are not directly relevant to this study. The accessible population also includes approximately 350 Form 4 students across the five schools, based on average class sizes of 70 students per school (some schools have two or three Form 4 streams). The student population is roughly balanced by gender, with estimates of 48 percent female and 52 percent male, reflecting the general demographics of secondary education in the North West Region.

Given the relatively small size of the accessible population (45 teachers, 350 students), the study aims to collect data from the entire accessible population rather than drawing a smaller sample. In research terminology, this is a census rather than a sample, which eliminates sampling error and increases the representativeness of the findings. However, practical constraints (teacher absenteeism, student consent refusals) may prevent the inclusion of all members of the accessible population, in which case the achieved sample will be described as a convenience sample drawn from the accessible population.

3.4 Sample and Sampling Technique

For this study, the sample consists of all teachers who teach Form 4 classes in the five selected secondary schools and all Form 4 students enrolled in those schools during the 2023/2024 academic year. The expected sample size is 40 teachers and 300 students (accounting for some absentees or refusals). Regarding the selection of the five schools, a purposive sampling technique is used. The researcher selects five schools that represent the diversity of school types in Bamenda II Subdivision. Specifically, the sample includes two public schools, two church-affiliated private schools, and one community-owned private school. This stratification ensures that the findings are not biased toward a single school type. Within each school, all Form 4 teachers and all Form 4 students are invited to participate, meaning that no further

sampling is required at the individual level. This approach is sometimes called total population sampling, and it maximizes the statistical power of the study.

For teachers, inclusion criteria are: (1) currently teaching at least one Form 4 class in the selected school; (2) teaching one of the core subjects (English, Mathematics, or Biology); and (3) having taught at that school for at least one full term prior to data collection, ensuring sufficient interaction with students. Exclusion criteria are teachers on extended leave, teachers who joined the school after the start of the academic term, and student teachers on internship. For students, inclusion criteria are: (1) currently enrolled in Form 4 in one of the five selected schools; (2) having been enrolled at that school since the beginning of the academic term; and (3) having parental or guardian consent to participate.

3.5 Instrumentation

Three primary instruments are used in this study. The first instrument is the **Teacher Characteristics Questionnaire (TCQ)**, designed for completion by teachers. The TCQ has three sections. Section A collects demographic information including age, gender, highest academic degree, professional teaching certification (type and year obtained), subject(s) taught, total years of teaching experience (both in current school and overall), and years of experience teaching the current subject. Section B consists of 20 Likert-scale items measuring teacher self-perceptions of content mastery (5 items, e.g., “I feel confident teaching advanced topics in my subject”), communication skills (5 items, e.g., “I explain concepts in multiple ways until students understand”), classroom management (5 items, e.g., “I have clear rules that students follow consistently”), and motivation (5 items, e.g., “I enjoy teaching even when students struggle”). Each item is rated on a 4-point scale: 1=Strongly Disagree, 2=Disagree, 3=Agree, 4=Strongly Agree. Section C contains three open-ended questions asking teachers to describe their strengths and weaknesses, the main challenges they face in teaching in Bamenda II, and their suggestions for improving student performance.

The second instrument is the **Student Perception of Teaching Questionnaire (SPTQ)**, designed for completion by Form 4 students. The SPTQ measures student perceptions of their teachers’ communication skills and classroom management behaviors. The instrument is shortened and simplified for student comprehension, using age-appropriate language and concrete examples. It includes 15 Likert-scale items: 8 items measuring communication skills (e.g., “My teacher explains lessons clearly,” “My teacher uses examples I can understand,” “My teacher asks if we have questions”) and 7 items measuring classroom management (e.g., “My teacher starts class on time,” “My teacher handles disruptions calmly,” “My teacher treats all students fairly”). Items are rated on a 3-point scale: 1=No, 2=Sometimes, 3=Yes. The SPTQ

also includes two open-ended questions: “What does your teacher do that helps you learn best?” and “What could your teacher do differently to help you learn better?”

The third instrument is the **Student Academic Performance Data Collection Sheet**, designed to be completed by school principals or records officers. This sheet requests the end-of-term examination scores for each participating Form 4 student in English Language, Mathematics, and Biology for the first and second terms of the 2023/2024 academic year. To protect student privacy, students are identified only by a code number, and the sheet does not request student names. For each student, the sheet also records age, gender, and attendance percentage (to be used as a control variable in analysis).

3.6 Validation and Reliability of Instruments

To ensure that the instruments measure what they intend to measure (validity) and produce consistent results (reliability), a multi-step process is followed. For content validity, the initial draft of the TCQ and SPTQ is reviewed by three experts: two faculty members from the University of Bamenda’s Department of Educational Foundations and one experienced secondary school principal from Bamenda (not involved in the main study). The experts are asked to assess whether each item is relevant, clear, and representative of the construct it is intended to measure. Based on their feedback, items that are ambiguous, irrelevant, or redundant are revised or removed.

For face validity, the revised instruments are piloted with a small group of five teachers and fifteen students from a school in Bamenda III Subdivision (adjacent to the study area but not included in the main study). Participants are asked to complete the questionnaire and then provide feedback on whether any items are confusing, offensive, or difficult to answer. They are also asked to time how long the questionnaire takes to complete. Based on this feedback, minor wording changes are made to improve clarity, particularly for student items.

For reliability, the internal consistency of the Likert-scale items in the TCQ and SPTQ is assessed using Cronbach’s alpha coefficient. During the pilot test, the pilot sample of five teachers and fifteen students completes the questionnaires twice, with a two-week gap between administrations. The data from the first administration are used to calculate Cronbach’s alpha for each subscale. Acceptable reliability is indicated by alpha values of 0.70 or higher. For the TCQ, the following subscales are expected: content mastery (5 items), communication skills (5 items), classroom management (5 items), and motivation (5 items). For the SPTQ, the subscales are communication skills (8 items) and classroom management (7 items). If the alpha values fall below 0.70, problematic items are identified and removed or rewritten. The test-

retest reliability is also calculated by correlating scores from the two administrations two weeks apart; a correlation of 0.80 or higher is considered evidence of good stability over time.

3.7 Data Collection Procedure

Data collection proceeds in four phases over an eight-week period. In Phase 1 (weeks 1-2), the researcher obtains ethical approval from the University of Bamenda's Institutional Review Board and obtains permission from the Divisional Delegate of Secondary Education for Mezam. The researcher then visits the five selected schools, meets with each principal, explains the purpose and procedures of the study, and obtains written permission to conduct research on the school premises. During these visits, the researcher also requests access to student records to obtain end-of-term examination scores and attendance data.

In Phase 2 (weeks 3-4), the researcher distributes parental consent forms. For each Form 4 student, the researcher provides a consent form to be signed by a parent or guardian. The consent form explains the study's purpose, the voluntary nature of participation, the confidentiality protections, and the fact that students may withdraw at any time without penalty. Only students with signed parental consent forms are allowed to participate. During this phase, the researcher also schedules dates and times for teacher and student questionnaire administration in coordination with each principal.

In Phase 3 (week 5), the researcher administers the questionnaires. Teachers complete the TCQ during a staff meeting or during free periods, with the researcher present to answer questions. Completion takes approximately 20-25 minutes. Students complete the SPTQ in their classrooms during a regular class period (with the subject teacher absent to avoid social desirability bias). The researcher reads each item aloud while students follow along and mark their answers, to accommodate students with varying reading levels. Completion takes approximately 15 minutes. Questionnaires are collected immediately upon completion.

In Phase 4 (week 6), the researcher collects student academic performance data. The researcher meets with each school's records officer and extracts end-of-term examination scores for English Language, Mathematics, and Biology for each participating student. Scores are recorded on the data collection sheet using student codes only. The researcher also collects attendance data (the percentage of days each student attended school during the two terms) to be used as a control variable.

Throughout the data collection process, the researcher maintains a field log, recording any observations, challenges, or informal conversations that may provide context for interpreting the quantitative results.

3.8 Data Analysis Technique

Data analysis proceeds in three stages: data cleaning, descriptive statistics, and inferential statistics. All analyses are performed using SPSS version 27. At the data cleaning stage, the researcher checks for missing data, outliers, and violations of statistical assumptions. Cases with more than 10 percent missing data are excluded from the analysis. Missing data on individual items are handled using mean imputation (replacing missing values with the mean for that item) if the missing data are random and limited. Outliers (extreme values more than 3 standard deviations from the mean) are examined; if they are clearly errors (e.g., a score of 150 percent), they are corrected or removed. Normality of the dependent variable (student performance) is assessed using the Kolmogorov-Smirnov test and by examining skewness and kurtosis values.

At the descriptive statistics stage, the researcher calculates frequencies, percentages, means, and standard deviations for all variables. For teacher characteristics, descriptive statistics are calculated by school and by subject. For student performance, mean scores (averaged across English, Mathematics, and Biology, and across two terms) are calculated for each student. The mean performance for the entire sample and for subgroups (by school, gender, teacher characteristics) are reported. Tables and charts are generated to visualize the data.

At the inferential statistics stage, the researcher tests the five hypotheses using a combination of statistical tests. First, Pearson's product-moment correlation (r) is calculated to examine the bivariate relationship between each teacher characteristic (qualifications, experience, content mastery, communication skills, classroom management) and student performance. The magnitude and direction of r indicate the strength of each relationship. Second, independent-samples t -tests are used to compare student performance between two groups (e.g., teachers with professional certification vs. without; teachers teaching in their specialization vs. out of specialization). Third, one-way ANOVA is used to compare student performance across three or more groups (e.g., teacher qualification levels: GCE A-Level only, HND, Bachelor's+PGDE, Master's; experience categories: 0-3, 4-7, 8-12, 13+ years).

Fourth, and most importantly, multiple linear regression is performed to determine the unique contribution of each teacher characteristic to student performance while controlling for potential confounding variables (student gender, student attendance, school type, class size). The regression model is specified as:

$$Y = \beta_0 + \beta_1(\text{Qualifications}) + \beta_2(\text{Experience}) + \beta_3(\text{Content Mastery}) + \beta_4(\text{Communication}) + \beta_5(\text{Classroom Management}) + \beta_6(\text{Attendance}) + \beta_7(\text{Gender}) + \beta_8(\text{School Type}) + \varepsilon$$

Where Y is student performance (average percentage score), β_0 is the intercept, β_1 through β_8 are regression coefficients representing the change in Y associated with a one-unit change in each independent variable holding others constant, and ε is the error term.

The regression analysis provides the most rigorous test of the study's hypotheses because it isolate the effect of each teacher characteristic. Statistical significance is set at $p < 0.05$. For qualitative data from open-ended questions, the researcher uses thematic analysis: reading all responses, generating initial codes (short labels describing the content), grouping codes into themes, and selecting illustrative quotes.

3.9 Ethical Considerations

This study adheres to the ethical principles outlined in the University of Bamenda's Research Ethics Policy, which is based on the Belmont Report (respect for persons, beneficence, and justice). First, **informed consent** is obtained from all participants. Teachers receive an information sheet explaining the study's purpose, procedures, risks (none beyond minor inconvenience), benefits (potential to improve teacher training), and the voluntary nature of participation. Teachers indicate consent by signing a consent form. For students, parental consent is obtained as described above, and students also provide verbal or written assent after having the study explained in age-appropriate language.

Second, **anonymity and confidentiality** are strictly protected. Student names are not recorded on any instrument; each student is assigned a code number, and the linking file (matching names to codes) is stored separately on a password-protected computer accessible only to the researcher. Teacher names are collected only on signed consent forms, which are stored separately from completed questionnaires. All data are reported in aggregate form, with no individual teacher or student identifiable in any publication or presentation. School names are replaced with pseudonyms (e.g., School A, School B).

Third, the principle of **do no harm** is observed. The study does not involve any experimental manipulation, deception, or invasive procedures. Teachers and students are not penalized in any way for declining to participate or for withdrawing after consenting. School administrators are not provided with individual responses, ensuring that teachers need not fear reprisal for honest self-assessments. The study does not collect any sensitive personal information (e.g., health, finances, political views).

Fourth, **voluntary participation and withdrawal** are emphasized. Participants are informed that they may skip any question they prefer not to answer and may stop participating at any

time without providing a reason. No coercion or undue influence (e.g., offering grades or money) is used.

Fifth, the study is **approved by relevant authorities**. Prior to data collection, the researcher obtains ethical approval from the University of Bamenda's Institutional Review Board (approval number: IRB-UB-2024-032). The researcher also obtains permission from the Divisional Delegate of Secondary Education for Mezam and from the principal of each participating school. All approvals are documented and presented upon request.

Finally, the researcher commits to **disseminating findings** to participating schools and to the Divisional Delegation of Secondary Education for Mezam. A plain-language summary of the results will be provided to each school, and a more detailed report will be shared with education officials to inform policy and practice. This commitment reflects the principle of reciprocity: participants contributed their time and information, and in return they deserve to benefit from the knowledge generated.

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